SA for Contraction Mappings

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Suppose we don't have access to $T(x)$, but only $T(x)+$noise. Then, we cannot do the fixed-point iteration to compute the fixed point. Instead, consider the following algorithm:

$$x_{k+1} = (1-\varepsilon_k)x_k + \varepsilon_k\left(T(x_k)+w_k\right),$$

where $\{w_k\}$ is i.i.d. noise and ε_k is called step-size or learning rate. For now, think of ε_k as a small number, we will discuss choices for ε_k later.

The above recursion is called stochastic approximation and is also written as

$$x_{k+1} = x_k - \varepsilon_k\left(x_k - T(x_k) - w_k\right)$$

We want to prove that this algorithm converges to a fixed point under the assumptions:

(a) T is a contraction mapping from $\mathbb{R}^n \to \mathbb{R}^n$ with contraction parameter α in the p^{th} norm

(b) $\{w_k\}$ satisfy $E(w_k \mid F_{k-1}) = 0$, $E(\|w_k\|_p^2 \mid F_{k-1}) \leq \sigma^2$, where $F_{k-1} = \{x_0, w_0, w_1, \ldots, w_{k-1}\}$

(c) $\{\varepsilon_k\}$ chosen suitably (explained later)

Define $\Delta_k = x_k - x^*$, where $x^* = T(x^*)$

$$\Rightarrow \quad \Delta_{k+1} \underset{\substack{\uparrow\\ \text{since } x^*=T(x^*)}}{=} \Delta_k - \varepsilon_k\left(\Delta_k - T(x_k) + T(x^*) - w_k\right)$$

Assume $2 \leq p < \infty$. Later we will extend the result to $p=\infty$.

Consider the Lyapunov function

$$W(\Delta) = \frac{1}{2}\|\Delta\|_p^2$$

(from now onwards, I may sometimes drop the subscript p, $\|\cdot\|$ should be understood $\|\cdot\|_p$ for this lecture)

We will use the following lemma, which we will prove later:

$$\boxed{W(\Delta_{k+1}) \leq W(\Delta_k) + \nabla^T W(\Delta_k)(\Delta_{k+1}-\Delta_k) + \frac{1}{2}(p-1)\|\Delta_{k+1}-\Delta_k\|_p^2}$$

Lemma

$$\Rightarrow W(\Delta_{k+1}) \leq W(\Delta_k) + \nabla^T W(\Delta_k)\Big(-\varepsilon_k\Delta_k + \varepsilon_k\big(T(x_k) - T(x^*)\big) + \varepsilon_k w_k\Big) + \frac{1}{2}(p-1)\Big\|-\varepsilon_k\Delta_k + \varepsilon_k\big(T(x_k) - T(x^*)\big) + w_k\Big\|_p^2$$

$$E\left(W(\Delta_{k+1}) - W(\Delta_k) \mid F_{k-1}\right)$$

$$= -\varepsilon_k \nabla^T W(\Delta_k)\Delta_k \quad — Ⓐ$$

$$+ \varepsilon_k \nabla^T W(\Delta_k)\left(T(x_k)-T(x^*)\right) \quad — Ⓑ$$

$$+ \varepsilon_k \nabla^T W(\Delta_k) \underbrace{E(w_k \mid F_{k-1})}_{=0}$$

$$+ \frac{1}{2}(p-1)\left[2\varepsilon_k^2\|T(x_k)-T(x^*)-\Delta_k\|^2 + 2\varepsilon_k^2 \underbrace{E\left(\|w_k\|_p^2 \mid F_{k-1}\right)}_{\leq \sigma^2}\right] \quad — Ⓒ$$

triangle inequality and $(a+b)^2 \leq 2(a^2+b^2)$

> To proceed further, note that $\nabla W(\Delta_k)$ is given by (to be proved later)
>
> $$\nabla W(\Delta_k) = G\|\Delta_k\|_p^{2-p},$$
>
> where $G(i) = |\Delta_k(i)|^{p-1}\operatorname{sgn}(\Delta_k(i))$
>
> and
>
> $$\nabla^T W(\Delta_k)\Delta_k = \|\Delta_k\|^2$$

$\Rightarrow$ Ⓐ is

$$-\varepsilon_k \nabla^T W(\Delta_k)\Delta_k = -\varepsilon_k\|\Delta_k\|^2$$

and

Ⓑ is

$$\varepsilon_k \nabla^T W(\Delta_k)\left(T(x_k)-T(x^*)\right)$$

$$= \varepsilon_k \|\Delta_k\|_p^{2-p} G^T\left(T(x_k)-T(x^*)\right),$$

where $G(i) = |\Delta_k(i)|^{p-1}\operatorname{sgn}(\Delta_k(i))$

$$\leq \varepsilon_k \|\Delta_k\|_p^{2-p}\|G\|_q\|T(x_k)-T(x^*)\|_p,$$

where we've used Holder's inequality $|a^Tb| \leq \|a\|_q\|b\|_p$ for p, q s.t. $\frac{1}{p}+\frac{1}{q}=1$ or $q = \frac{p}{p-1}$

Since $\|T(x)-T(x^*)\|_p \leq \alpha\|x_k - x^*\|_p = \alpha\|\Delta\|_p$

and

$$\|G\|_q = \left(\sum_{i=1}^n |\Delta_k(i)|^{(p-1)q}\right)^{1/q} = \left(\sum_{i=1}^n |\Delta_k(i)|^p\right)^{\frac{p-1}{p}} = \|\Delta_k\|_p^{p-1}$$

$$\Rightarrow Ⓑ \leq \alpha\varepsilon_k\|\Delta_k\|_p^{2-p}\|\Delta_k\|^{p-1}\|\Delta_k\|_p = \alpha\varepsilon_k\|\Delta_k\|_p^2$$

Next,

$$Ⓒ = (p-1)\left(\|-\varepsilon_k\Delta_k + \varepsilon_k(T(x_k)-T(x^*))\|_p^2 + \varepsilon_k^2\sigma^2\right)$$

$$\leq (p-1)\left(\left(\|\varepsilon_k\Delta_k\| + \varepsilon_k\|T(x_k)-T(x^*)\|\right)^2 + \varepsilon_k^2\sigma^2\right)$$

↑ Triangle inequality

$$\leq (p-1)\left(2\varepsilon_k^2\|\Delta_k\|^2 + 2\varepsilon_k^2\alpha^2\|\Delta_k\|^2 + \varepsilon_k^2\sigma^2\right)$$

↑ $(a+b)^2 \leq 2(a^2+b^2)$ and T is a p-norm contraction

$$= 2(p-1)\left(\varepsilon_k^2(1+\alpha^2)\|\Delta_k\|^2 + \varepsilon_k^2\sigma^2\right)$$

Putting it all together,

$$E\left(W(\Delta_{k+1}) - W(\Delta_k) \mid x_k\right) \leq \left(-\varepsilon_k + \alpha\varepsilon_k + (p-1)\varepsilon_k^2(1+\alpha^2)\right)\underbrace{\|\Delta_k\|^2}_{=2W(\Delta_k)} + \varepsilon_k^2\sigma^2$$

$$\Rightarrow E(W(\Delta_{k+1})) \leq \left(1-2\varepsilon_k(1-\alpha) + 2(p-1)(1+\alpha^2)\varepsilon_k^2\right)E(W(\Delta_k)) + \varepsilon_k^2\sigma^2$$

Suppose $\varepsilon_k = \varepsilon$ (other, more interesting choices of ε_k will be explored in the homework):

then, if ε is sufficiently small,

$$\underbrace{1-2\varepsilon(1-\alpha)+2(p-1)(1+\alpha^2)\varepsilon^2}_{\triangleq \beta} \in [0,1]$$

$$\Rightarrow E(W(\Delta_{k+1})) \leq \beta E(W(\Delta_k)) + \varepsilon^2\sigma^2$$

$$\Rightarrow E(W(\Delta_k)) \leq \beta^k E(W(\Delta_0)) + \varepsilon^2\sigma^2\underbrace{\left(\beta^{k-1}+\beta^k+\cdots+1\right)}_{\leq \frac{1}{1-\beta}}$$

$$\Rightarrow \boxed{\frac{1}{2}E\left(\|\Delta_k\|_p^2\right) \leq \frac{\beta^k}{2}E\left(\|\Delta_0\|_p^2\right) + \frac{\varepsilon^2\sigma^2}{1-\beta}}$$

↑ finite-time bound

$$\boxed{\lim_{k\to\infty}\frac{1}{2}E\left(\|\Delta_k\|_p^2\right) \leq \frac{\varepsilon\sigma^2}{2(1-\alpha)-2(p-1)(1+\alpha^2)\varepsilon}}$$

↳ Asymptotic bound

Proof of the lemma:

Let $f(x) = \frac{1}{2}\|x\|_p^2$

By Taylor's theorem,

$$f(y) = f(x) + \nabla^T f(x)(y-x) + (y-x)^T\nabla^2 f(z)(y-x)$$

for some $z = \alpha x + (1-\alpha)y$ where $\alpha \in [0,1]$

As we will show later,

$$\frac{\partial f}{\partial x(i)} = \|x\|^{2-p}|x(i)|^{p-1}\operatorname{sgn}(x(i))$$

$$\Rightarrow \frac{\partial^2 f}{\partial^2 x(i)} = \|x\|^{2-p}(p-1)|x(i)|^{p-2} + |x(i)|^{p-1}\cancel{\operatorname{sgn}(x(i))}(2-p)\|x\|^{1-p}\frac{1}{\cancel{p}}\left(\sum_i |x(i)|^p\right)^{\frac{1}{p}-1}\cancel{p}|x(i)|^{p-1}\cancel{\operatorname{sgn}(x(i))}$$

$$= \|x\|^{2-p}(p-1)|x(i)|^{p-2} + (2-p)|x(i)|^{2(p-1)}\|x\|^{2(1-p)}$$

and if $j \neq i$,

$$\frac{\partial^2 f}{\partial x(i)\partial x(j)} = |x(i)|^{p-1}\operatorname{sgn}(x(i))(2-p)\|x\|^{2-2p}|x(j)|^{p-1}\operatorname{sgn}(x(j))$$

$\Rightarrow$ letting $d = y-x$,

$$d^T\nabla^2 f(x)d = \sum_{\substack{i,j\\ i\neq j}} d_i\left(\nabla^2 f(z)\right)_{ij}d_j + \sum_i d_i^2\left(\nabla^2 f(z)\right)_{ii}$$

$$= (2-p)\|z\|^{2-2p}\left(\sum_i |z(i)|^{p-1}\operatorname{sgn}(z(i))d_i\right)^2 + \|z\|^{2-p}\sum_i (p-1)|z(i)|^{p-2}d_i^2$$

$$\underset{\substack{\uparrow\\ \text{since } 2-p\leq 0}}{\leq} (p-1)\sum_i \frac{|z(i)|^{p-2}}{\|z\|^{p-2}}d_i^2 \leq \frac{(p-1)}{\|z\|^{p-2}}\left(\sum_i\left(|z(i)|^{p-2}\right)^{\frac{p}{p-2}}\right)^{\frac{p-2}{p}}\underbrace{\left(\sum_i\left(d_i^2\right)^{\frac{p}{2}}\right)^{\frac{2}{p}}}_{\|d\|_p^2}$$

$$= \frac{(p-1)}{\cancel{\|z\|^{p-2}}}\cancel{\|z\|_p^{p-2}}\|d\|_p^2$$

$$= (p-1)\|d\|_p^2$$

□

Finally, let's consider the case where T is an $\|\cdot\|_\infty$ contraction. Consider some $p<\infty$, then

$$\|T(x_1)-T(x_2)\|_p^p = \sum_{i=1}^n |T(x_1)-T(x_2)|_i^p \leq n\max_i |T(x_1)-T(x_2)|_i^p = n\|T(x_1)-T(x_2)\|_\infty^p$$

$$\Rightarrow \|T(x_1)-T(x_2)\|_p \leq n^{1/p}\|T(x_1)-T(x_2)\|_\infty \leq n^{1/p}\alpha\|x_1-x_2\|_\infty \leq n^{1/p}\alpha\|x_1-x_2\|_p$$

If p is large, $n^{1/p}\alpha < 1$ and thus T is also a contraction in the p-norm for sufficiently large p.

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Gradient of W:

$$W(\Delta_k) = \frac{1}{2}\|\Delta_k\|_p^2 = \frac{1}{2}\left(|\Delta_k(1)|^p + \cdots + |\Delta_k(n)|^p\right)^{\frac{2}{p}}$$

↑ assuming $\Delta \in \mathbb{R}^n$

$$\frac{\partial W(\Delta)}{\partial \Delta_i} = \frac{\cancel{2}}{\cancel{2}\cancel{p}}\left(|\Delta_k(1)|^p + \cdots + |\Delta_k(n)|^p\right)^{\frac{2}{p}-1}\cancel{p}\operatorname{sgn}(\Delta_k(i))|\Delta_k(i)|^{p-1}$$

$$= \|\Delta_k\|_p^{2-p}\operatorname{sgn}(\Delta_k(i))|\Delta_k(i)|^{p-1},$$

where $\operatorname{sgn}(\Delta)=0$ if $\Delta=0$

$$= \|\Delta_k\|_p^{2-p}|\Delta_k(i)|^{p-1}\operatorname{sgn}(\Delta_k(i))$$

$$\Rightarrow \nabla W(\Delta_k) = G\|\Delta_k\|_p^{2-p},$$

where $G(i) = |\Delta_k(i)|^{p-1}\operatorname{sgn}(\Delta_k(i))$

Further,

$$\nabla^T W(\Delta_k)\Delta_k = \sum_{i=1}^n \frac{\partial W(\Delta_k)}{\partial \Delta_i}\Delta_k(i)$$

$$= \|\Delta_k\|_p^{2-p}\sum_{i=1}^n |\Delta_k(i)|^{p-1}\underbrace{\operatorname{sgn}(\Delta_k(i))\Delta_k(i)}_{|\Delta_k(i)|}$$

$$= \|\Delta_k\|_p^{2-p}\|\Delta_k\|_p^p$$

$$= \|\Delta_k\|^2$$

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